A Capstone Document

Professional Certification – Data Science

Electricity Price Prediction

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I. Introduction

Accurately forecasting electricity prices and demand is a complex challenge due to the many dynamic factors that influence the energy market. Variables such as weather conditions, time of day, seasonal patterns, and public holidays all contribute to fluctuations in electricity consumption and pricing. These rapid and often unpredictable changes make it difficult for energy providers, market participants, and consumers to plan effectively.

This project addresses this challenge by analyzing historical electricity market data and developing a machine learning based predictive system capable of forecasting electricity prices, system load, and wind energy production with improved accuracy. By uncovering underlying patterns and trends within the data, the model aims to provide deeper insights into how external factors affect market behavior.

II. Goals and Objectives

The main goal of this project is to analyze the historical electricity market data and build a predictive machine or system that will forecast electricity prices and demand with high accuracy.

- To understand patterns and trends for electricity prices and demand.
- To examine how external factors like holidays, seasonal changes affect the electricity prices and system load
- To develop a machine learning model that could better predict electricity prices, system load and wind energy production.

III. Dataset Overview

This section outlines the key attributes of the dataset and their relevance to the predictive modeling process.

a. About the Dataset

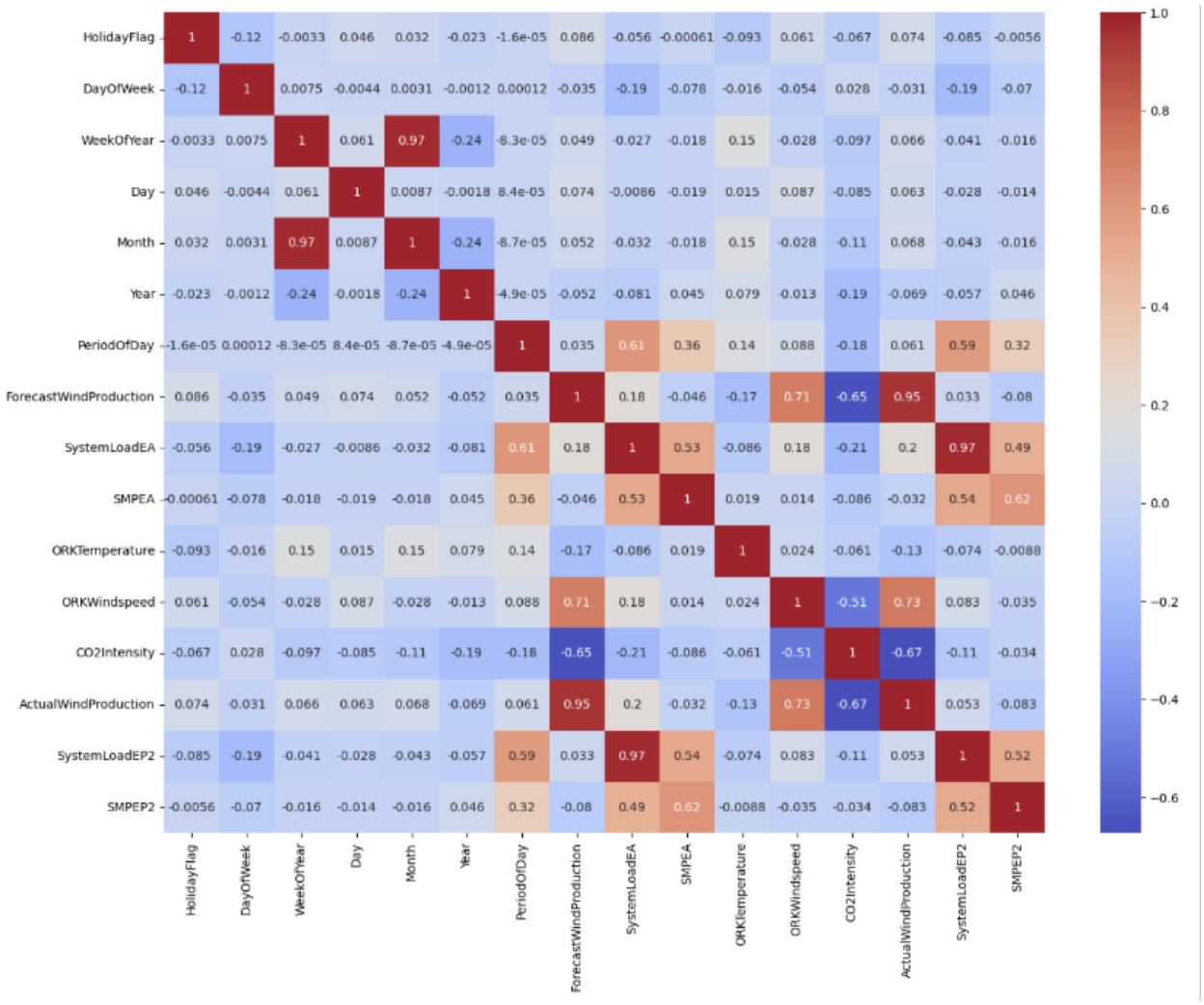
This project showcases the historical electricity market information that helps uncover seasonal trends, patterns and demand fluctuations in the electricity grid. It contains 18 features with a total of 38,014 records which is very important to train the model to get more accurate results.

b. Features

The dataset includes a range of features that provides important context for understanding market behavior and supports the development of an accurate predictive model.

1. DateTime - This is the exact date and time of when the electricity data was recorded.
2. Holiday - This feature contains the name of the holiday if it is, and returns “none” if not.
3. HolidayFlag - This indicates whether the date is a holiday (1) or not (0).
4. DayOfWeek - This is a numeric representation of the day (0-6), 0 being Monday.
5. WeekOfYear - This contains the week number within the year (1-52).
6. Day - The day of the month the data was recorded (1-31).
7. Month - The month the data was recorded (1-12).
8. Year - The year when the data was recorded.
9. PeriodOfDay - This represents a time period of the day (30 minute segment) when the data was recorded.
10. ForecastWindProduction - The forecasted amount of electricity expected to be generated from wind power.
11. SystemLoadEA - The forecasted electricity demand when the data was recorded.
12. SMPEA (Single Market Price for Electricity) - This is the forecasted electricity price.
13. ORKTemperature - This is the actual temperature reading of the time and location the data was recorded.
14. ORKWindspeed - This is the actual wind speed reading of the time and location the data was recorded.
15. CO2Intensity - This is the actual C02 (carbon dioxide) for the electricity produced.
16. ActualWindProduction - The actual amount of electricity produced from the wind power.
17. SystemLoadEP2 - This is the actual electricity demand of the system.
18. SMPEP2 [TARGET]- This is the actual price of the consumed electricity and is the target of the machine learning model.

c. Correlation



To gain a deeper understanding of the variables used in this model, we analyzed their relationships through a correlation heatmap. This visualization helps us identify which features move together, how strongly they interact, and which ones may play a more significant role in predicting the target variable.

The heatmap reveals several strong positive correlations. The most notable ones appear between the forecasted electricity demand and the actual electricity demand of the system, indicating that the recorded forecasts align closely with real demand. Similarly, forecasted wind production shows a very strong positive relationship with actual wind production, suggesting consistent and reliable forecasting for wind energy. We can also observe strong correlations between the period-of-day–related variables and the system load features (SystemLoadEA and SystemLoadEP2), which highlights how energy demand varies throughout the day. And there's a positive correlation between ORKWindspeed and the variables of Wind Production.

In contrast, two strong negative correlations stand out. CO_2 intensity is strongly and inversely related to both forecasted and actual wind production. This makes intuitive sense, as higher wind production reduces reliance on fossil fuels and consequently lowers CO_2 emissions.

IV. Methodology

This section outlines the key steps of the modeling process to ensure that the final predictions are accurate, reliable, and well-supported by the data.

a. Machine Learning Model

In this project, we are using the Supervised - Regression approach since we are using historical data to make predictions on electricity prices and that the target variable is continuous. The model is going to be trained on a labeled data for it to learn patterns and make accurate predictions on unseen records.

According to the correlation analysis we considered these features have ore relevance for the model: "Day", "Month", "ForecastWindProduction", "SystemLoadEA", "SMPEA", "ORKTemperature", "ORKWindspeed", "CO2Intensity", "ActualWindProduction", "SystemLoadEP2" as our independent variables and “SMPEP2” as our target.

b. Wrangling

Data preprocessing was performed to ensure the dataset was clean, and suitable for building a reliable prediction model. Since most features were already well structured, only minimal wrangling steps were performed. Key transformations included:

- Replace missing values in “ForecastWindProduction”, “SMPEA”, “ActualWindProduction” and “SMPEP2” with its median as their distribution is skewed,
- Replace missing values in "SystemLoadEA", "ORKTemperature", "ORKWindspeed", "CO2Intensity", "SystemLoadEP2" with its mean as they have a normal distribution.
- Standard Scalar for DayOfWeek, WeekOfYear, Day, Month, Year, PeriodOfDay as they are time/calendar variables and they need standardization.

c. Cross-Validation

We’ve chosen 80% Train and 20% test for the cross validation to make sure that the model is properly trained to provide better and higher predictive accuracy.

d. Algorithms

When building a regression model, it is standard to test multiple algorithms because each one captures patterns differently.

- RandomForestRegressor: Random Forest is a powerful ensemble algorithm that handles non-linear relationships, interactions between variables, and noisy data extremely well. Electricity price data contains spikes and irregular fluctuations, which Random Forest can smooth out while still capturing major trends.
- LinearSVR: SVR is often tested because it performs well on datasets where relationships are not perfectly linear but still somewhat structured. It tries to fit the best margin around the data points.
- KNeighborsRegressor: It predicts by comparing a data point to its closest neighbors. Testing KNN helps understand whether similar historical conditions lead to similar price outcomes.
- LinearRegression: Linear Regression is a standard baseline model. It checks whether electricity prices can be explained through simple linear relationships between features such as system load, temperature, and wind production.
- DummyRegressor: A Dummy Regressor provides a baseline comparison by predicting something simple, like the mean or median.

e. Tools

To complete this project, several tools were used. The dataset was stored in local machine in csv form. To compare the same workflow across different platforms, the data wrangling steps and model creation were performed both in MLOS and in Python. The user interface was also developed in both environments, although the Python version was preferred due to its flexibility and clarity.

V. Machine Learning Result

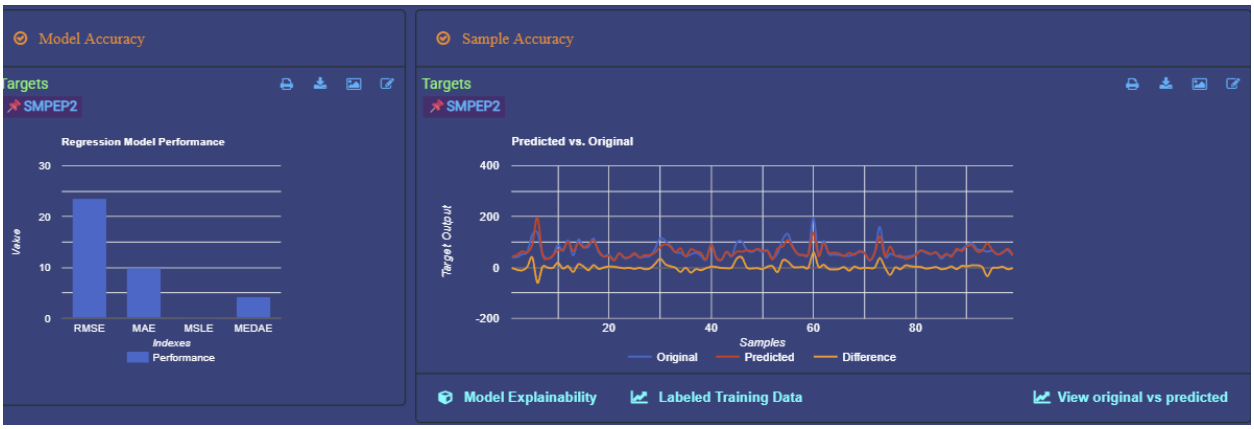
This section presents the performance of the machine learning model, highlighting key findings from the overall result analysis and evaluating how well the model predicts electricity prices and system load.

a. Overall Analysis

Once all the algorithms were applied, is possible to conclude that the one with a better performance was RandomForestRegressor. It’s erros is the smallest one (9.85) between the 5 versions of the model.

Version-Tag	Dataset	Algorithm	Rank	Error	Doc.	Publish	Delete
v.1-v.a02	updatedelectric-model1	RandomForestRegressor	1	9.85			
v.8-v.be0	updatedelectric-model1	LinearSVR	2	14			
v.7-v.897	updatedelectric-model1	KNeighborsRegressor	3	14.53			
v.2-v.876	updatedelectric-model1	LinearRegression	4	14.61			
v.3-v.860	updatedelectric-model1	DummyRegressor	5	21.38			

The results found with this algorithm show the chart on the right where is visible the model's predictions compared to actual electricity prices across several samples.



The blue line (original values) and the red line (predicted values) follow similar patterns. This indicates that the model successfully captures the general shape and direction of the target variable over time.

The yellow line represents the difference between predicted and actual values. It remains relatively close to zero for most samples, meaning the model is accurate on average. However, the error spikes slightly in areas where the original values show sharp increases or decreases. This is a common limitation for models like Random Forest, which tend to smooth extreme variations.

The bar chart on the left summarizes the model's performance using RMSE, MAE, MSLE, and MEDAE. These metrics provide different perspectives on prediction error.

RMSE (Root Mean Squared Error = 23.5) shows the average size of the error, with larger mistakes being penalized more heavily. The relatively high RMSE value indicates that the model experiences some larger errors typically when the target variable experiences sudden peaks or drops.

MAE (Mean Absolute Error = 9.85) is lower than the RMSE, suggesting that most errors are moderate and consistent. This means the model performs well under normal conditions but occasionally misses sharper fluctuations.

MSLE (Mean Squared Logarithmic Error = 0.05) value is very low, showing that the model handles proportional changes well. It indicates that the model does not dramatically mispredict low or mid-range values and performs reliably across different scales

MEDAE (Median Absolute Error = 4.35) demonstrates that the model's *typical* prediction error is minimal, and most predictions fall close to the true value. This confirms stable and reliable performance for most samples.

VI. User Interface Outcome

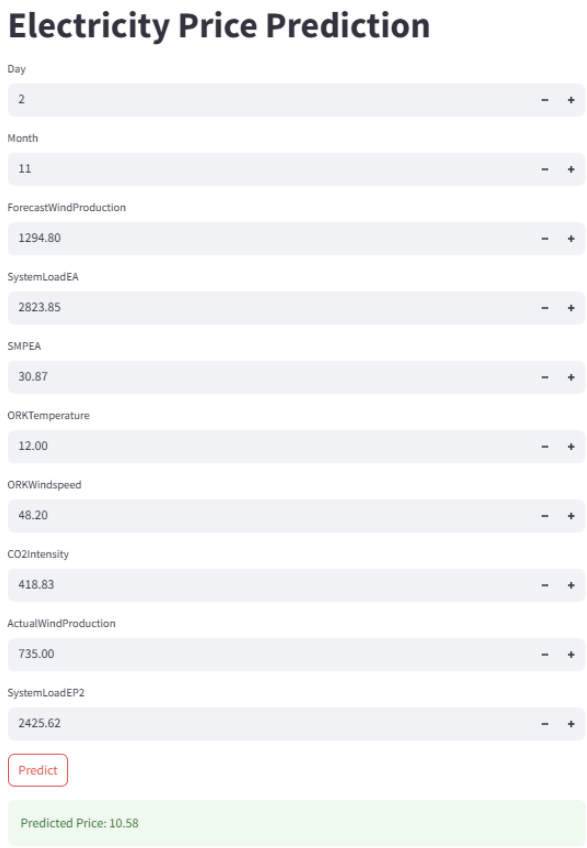
This section highlights the design and functionality of the applications developed for this project. It covers web application which offers access to online platforms.

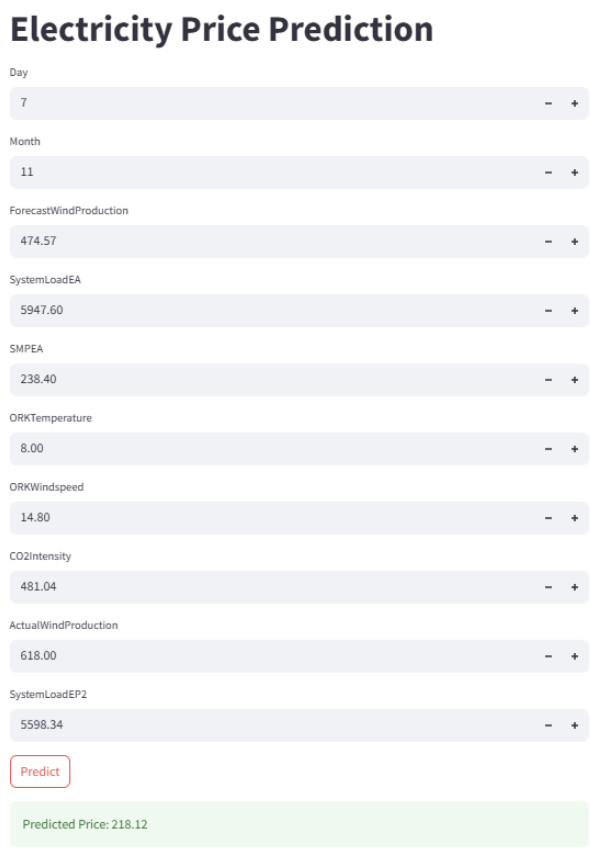
a. Web Application

To make the machine learning model accessible and easy to use, a simple and interactive user interface (UI) was developed using Streamlit, a Python framework designed for building data-driven web applications. The goal was to allow end-users to input relevant parameters, run predictions, and visualize results without needing to interact with the underlying code.

First, after the model was created and trained, it was saved using the pickle library. Inside the Streamlit app, the model is loaded at the beginning so it can be used when predictions are requested. The UI shows a title, and the input components. Each field is made to allow users to provide data to the model.

After collecting inputs, the app prepares them in the required format (usually a list or NumPy array) and sends them to the model. Streamlit's button triggers the prediction process and displays the result clearly on the interface.





In the first scenario, where system load values and wind-related features were moderate, the predicted electricity price was relatively low (10.58). In contrast, the second scenario included significantly higher system load values and higher energy demand indicators, resulting in a substantially higher predicted price (218.12). This behavior is consistent with real electricity market dynamics, where price increases are typically driven by higher demand and less favorable supply conditions.

VII. Conclusion

This project successfully explored historical electricity market data and demonstrated that machine learning can be an effective tool for predicting electricity prices. By analyzing 38,014 rows of time-based, environmental, and market-related features, we identified clear relationships among system load, wind production, weather conditions, and electricity pricing. The correlation analysis showed that factors such as system load and wind energy production have strong influence on actual market price , reinforcing the importance of both demand and renewable energy availability in price forecasting.

Overall, this project achieved its goals by:

- Understanding historical patterns and trends in electricity prices and system load
- Evaluating how holidays, seasonal changes, and weather factors influence price fluctuations
- Developing a machine learning model capable of predicting electricity prices with strong accuracy
- Demonstrating that market forecasting can support better planning, cost reduction, and more efficient energy management

But faced some limitations. Although the dataset contains 18 features, it still excludes several real-world factors that influence electricity prices, such as fuel costs, market regulations, transmission constraints, and sudden outages. These missing variables may reduce the model's ability to generalize to more complex market conditions.

Holiday impacts were represented only as a binary flag, and weather variables only included temperature and windspeed. Real impacts can vary based on location, severity, and daily behavioral changes, which are not fully captured.

Wind production is included, but the dataset does not consider solar power, hydro conditions, or fuel availability. Renewable energy fluctuations are complex and not fully represented.